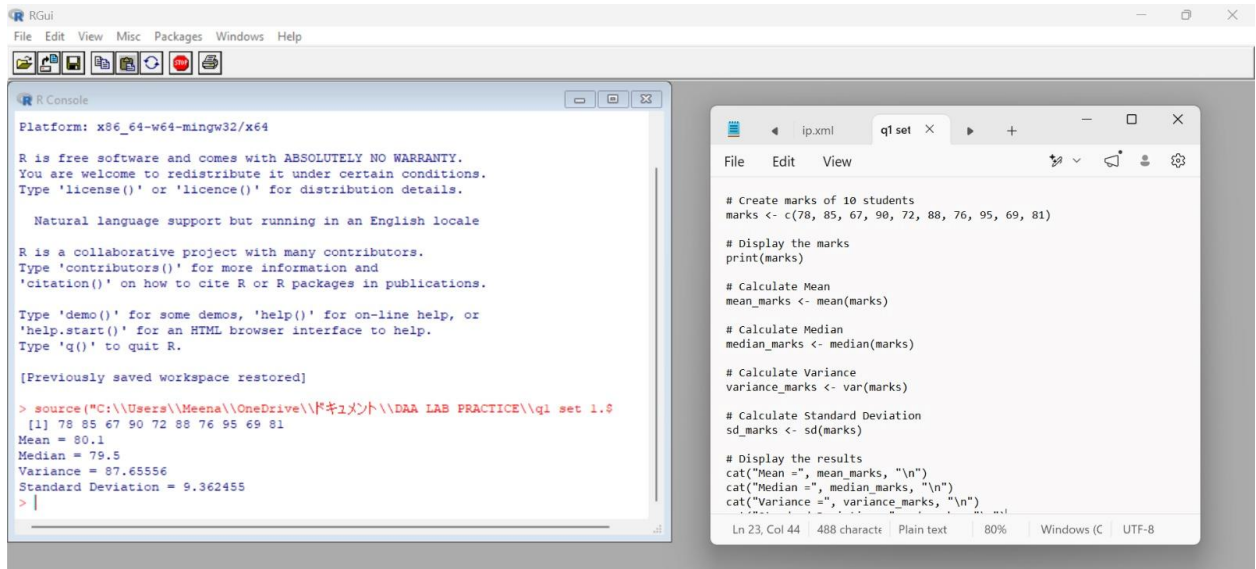


SET 1

1. R: Create 10 student marks and calculate mean, median, variance and standard deviation.

Output:



The screenshot shows the RGui interface. The R Console on the left displays the following output:

```
Platform: x86_64-w64-mingw32/x64
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]
> source("C:\\Users\\Meena\\OneDrive\\ドキュメント\\DAA LAB PRACTICE\\q1 set 1.R")
[1] 78 85 67 90 72 88 76 95 69 81
Mean = 80.1
Median = 79.5
Variance = 87.65556
Standard Deviation = 9.362455
> |
```

The script file 'q1 set 1.R' on the right contains the following code:

```
# Create marks of 10 students
marks <- c(78, 85, 67, 90, 72, 88, 76, 95, 69, 81)

# Display the marks
print(marks)

# Calculate Mean
mean_marks <- mean(marks)

# Calculate Median
median_marks <- median(marks)

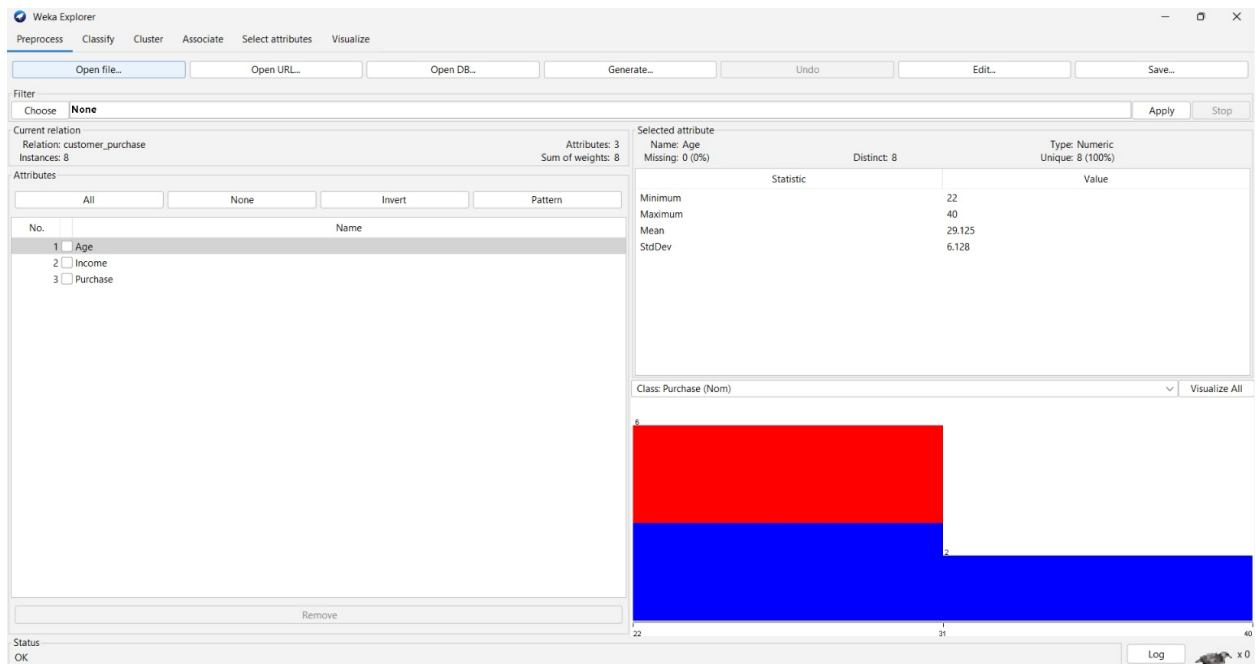
# Calculate Variance
variance_marks <- var(marks)

# Calculate Standard Deviation
sd_marks <- sd(marks)

# Display the results
cat("Mean =", mean_marks, "\n")
cat("Median =", median_marks, "\n")
cat("Variance =", variance_marks, "\n")
cat("Standard Deviation =", sd_marks, "\n")
```

2. R: For 12,18,21,25,29,31,35,40,44,50 find Q1,Q2,Q3,IQR and draw a boxplot.

Output:

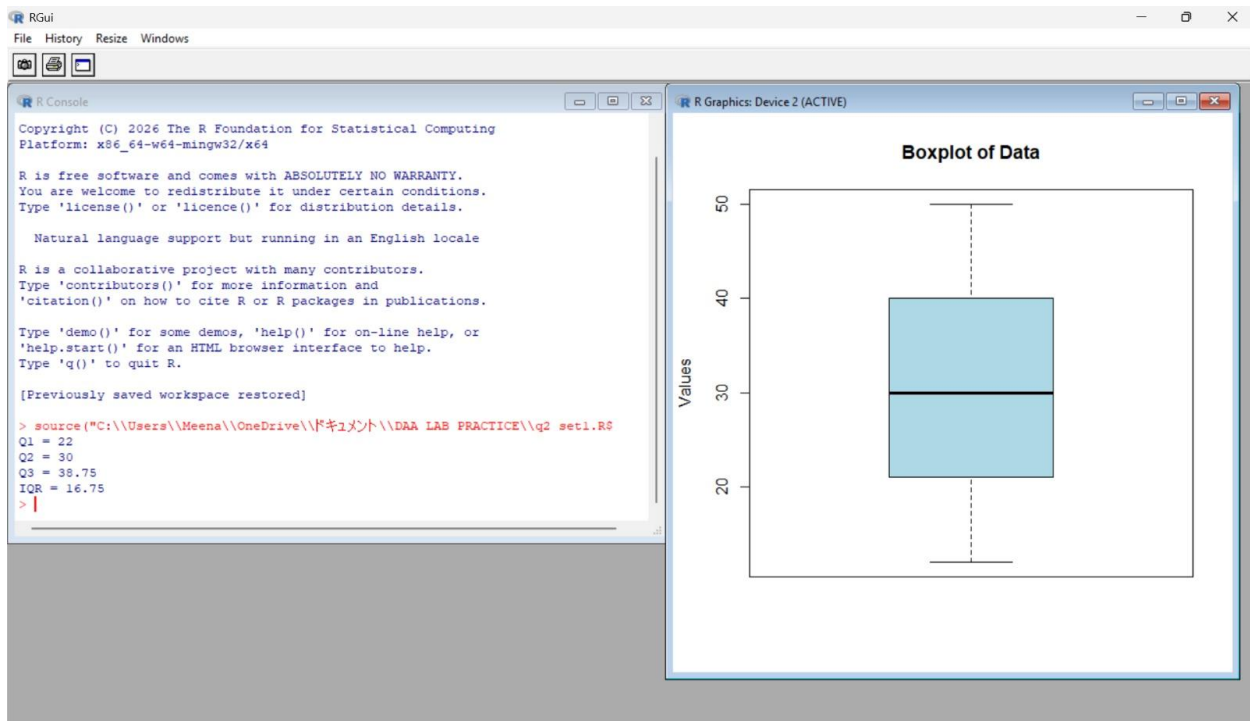


3. WEKA: Create an ARFF customer dataset (Age, Income, Purchase) and apply J48; visualize the tree and record correct/incorrect instances.

Dataset: Customer Purchase

```
@relation customer_purchase
@attribute Age numeric
@attribute Income numeric
@attribute Purchase {Yes,No}
@data
22,25000,No
25,32000,No
28,45000,Yes
31,52000,Yes
35,60000,Yes
40,72000,Yes
23,28000,No
29,48000,Yes
```

Output:



4. WEKA: Apply Naive Bayes to a suitable nominal dataset and report accuracy, precision, recall and F1-score.

Use the same Customer Purchase ARFF dataset given for Set 1 Q3.

Output:

The screenshot shows the Weka Explorer application window. The 'Classify' tab is active, and the 'NaiveBayes' classifier is selected. The 'Test options' section shows 'Cross-validation' with 'Folds' set to 10. The 'Classifier output' pane displays the following information:

```
=== Run information ===  
  
Scheme:      weka.classifiers.bayes.NaiveBayes  
Relation:     customer_purchase  
Instances:    0  
Attributes:   3  
              Age  
              Income  
              Purchase  
Test mode:    10-fold cross-validation  
  
=== Classifier model (full training set) ===  
  
Naive Bayes Classifier  
  
Attribute      Class  
                Yes      No  
                (0.6)    (0.4)  
=====
```

	Age	Income
mean	32.9143	24
std. dev.	4.9862	1.2122
weight sum	5	3
precision	2.5714	2.5714

```
Income  
mean      56400 29095.2381  
std. dev. 10049.0227 3165.1446  
weight sum      5      3  
precision  6714.2057 6714.2057  
  
Time taken to build model: 0 seconds
```

The status bar at the bottom indicates 'Problem evaluating classifier'.